

PAPER_375 — UQFF Advanced Integration

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Wormhole-MUGE Term | Meissner Exponential | Relativistic γ | Error Propagation

Star Magic UQFF Whitepaper Series

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Source Documents: "Compressed UQFF Equation_14May2025.docx",

"Master UQFF Resonance Equation_14May2025.docx",

"UQFFResonance Superconductive Universal Gravity Equation system proof set.15May2025.docx"

Abstract

This paper presents the advanced integration of the UQFF framework combining four new mathematical formulations: (1) a wormhole-MUGE coupling term derived from the Morris-Thorne metric; (2) the exponential Meissner superconductivity model replacing the linear B/Bcrit quenching; (3) a special-relativistic Lorentz correction applied to the DPM acceleration term for high-velocity systems; and (4) a formal error propagation formalism for uncertainty quantification across all MUGE terms. These four enhancements are combined into a single Unified UQFF master equation and validated for the J1610+1811 system at $v=0.99c$.

1. Wormhole-MUGE Coupling Term

The Morris-Thorne wormhole geometry (PAPER_373) introduces a new acceleration term in the MUGE resonance sum:

$$a_{\mathrm{worm}}(r) = \frac{f_{\mathrm{worm}} \cdot E_{\mathrm{vac,neb}}}{b^2 + r^2}$$

where:

$$f_{\mathrm{worm}} = 10^{-10} \text{ (dimensionless wormhole coupling constant)}$$

$$E_{\mathrm{vac,neb}} = 7.09 \times 10^{-36} \text{ J (nebulary vacuum energy)}$$

$$b = 1.0 \text{ m (wormhole throat radius)}$$

$$r = \text{evaluation radius (m)}$$

This term encodes the gravitational contribution of a wormhole throat at distance r , modulated by the vacuum energy of the local medium.

At $r = 1 \text{ m}$, $b = 1$:

$$a_{\mathrm{worm}}(1) = \frac{10^{-10} \times 7.09 \times 10^{-36}}{1 + 1} = 3.545 \times 10^{-46} \text{ m/s}^2$$

2. Meissner Exponential Superconductivity

PAPER_372 uses the linear Meissner approximation: $(1 - B/B_{\mathrm{crit}})$.

This paper introduces the physically more accurate **exponential form** applicable to Type-II superconductors (London penetration depth model):

$$\left(1 - \frac{B}{B_{\mathrm{crit}}}\right) \longrightarrow e^{-B/B_{\mathrm{crit}}}$$

For the Compressed UQFF master equation, the gravitational coupling becomes:

$$g_{\mathrm{UQFF}} = \frac{GM}{r^2} \cdot [1 + H_0 t] \cdot e^{-B/B_{\mathrm{crit}}} \cdot [1 + F_{\mathrm{env}}] + \dots$$

The exponential form ensures monotone decay from $e^0 = 1.0$ (no field) to 0 (field well above B_{crit}), without unphysical negative values that arise from the linear form when $B > B_{\mathrm{crit}}$.

System	B/B_{crit}	Linear factor	Exponential factor
SGR1745-2900	0.1	0.9	0.905
SgrA*	0.1	0.9	0.905
Student's Guide	0.1	0.9	0.905

3. Relativistic Lorentz Correction

For high-velocity systems (e.g., J1610+1811 jet at $v = 0.99c$), the DPM acceleration term undergoes Lorentz suppression:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$a_{\mathrm{DPM}} \longrightarrow \frac{a_{\mathrm{DPM}}}{\gamma}$$

For $v = 0.99c$:

$$\gamma = \frac{1}{\sqrt{1 - 0.9801}} = \frac{1}{\sqrt{0.0199}} \approx 7.089$$

This suppresses a_{DPM} by factor ~ 7 , reflecting that the DPM force (electromagnetic in origin) is frame-dependent in the relativistic regime. All other resonance terms retain their coordinate-frame values.

4. Error Propagation Formalism

Uncertainties in individual MUGE terms propagate in quadrature:

$$\Delta g = \sqrt{\sum_i (\Delta a_i)^2}$$

where Δa_i is the uncertainty in each term a_i . For a fractional error $f = 1\%$:

$$\Delta a_i = f \cdot |a_i|$$

$$\Leftrightarrow$$

$$\Delta g = f \cdot \sqrt{\sum_i a_i^2}$$

This provides a rigorous uncertainty bound for UQFF predictions, enabling comparison with observational error bars.

5. Unified UQFF Master Equation (Complete Form)

Combining all prior papers (PAPER_371–375):

$$\begin{aligned} g(r,t) = & \underbrace{\left[\frac{GM}{r^2} (1 + H_0 t) \right]}_{\text{e}^{-B/B_{\text{crit}}}} (1 + F_{\text{env}}) \\ & + \sum U_{\text{gi}} + \frac{\Lambda c^2}{3} + \frac{\hbar}{\Delta x \Delta p} \int \psi^* \hat{H} \psi \, dV \cdot \frac{2\pi}{t_H} \\ & + \rho_f v g + (M_{\text{vis}} + M_{\text{DM}}) \left[\frac{\delta \rho}{\rho} + \frac{3GM}{r^3} \right] \right]_{\text{Compressed UQFF (PAPER 372, Meissner exp)}} \\ & + \underbrace{\left[\frac{a_{\text{DPM}}}{\gamma} + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super,freq}} + a_{\text{aether,res}} \right]}_{\text{MUGE Resonance with Lorentz correction (PAPER 371)}} \\ & + U_{\text{g4i}} + a_{\text{quantum,freq}} + a_{\text{Aether,freq}} + a_{\text{fluid,freq}} \\ & + a_{\text{osc}} + a_{\text{exp,freq}} + f_{\text{TRZ}} \right]_{\text{Wormhole (PAPER 373)}} \\ & \pm \delta g \end{aligned}$$

6. Canonical Parameter Summary

Symbol	Value	Paper
f_{worm}	1×10^{-10}	PAPER_375
Meissner form	$\exp(-B/B_{\text{crit}})$	PAPER375 (vs linear PAPER372)
γ ($v=0.99c$)	≈ 7.089	PAPER_375
δg (1% error, SGR1745)	$\sim 10^{\blacksquare} \times 0.01$	PAPER_375

7. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace UQFFAdvanced

compute_a_wormhole(Evac_neb, b, r) — wormhole MUGE term
meissner_exp(B, Bcrit) — exponential Meissner factor
lorentz_gamma(v) — relativistic Lorentz factor
apply_lorentz(aDPM, v) — Lorentz-corrected DPM
error_propagation(delta_terms) — quadrature error propagation
compute_unified_UQFF(sys, res, t, v_jet, b_worm, r_worm) — master unified function
compute_total_uncertainty(sys, p, frac_error) — uncertainty budget

Python: CondensedPhysics4.py, class UQFFWormholeMeissnerRelativisticGammaCalculator (CP4 #23)

WOLFRAM_TERM: WOLFRAM_TERM_UQFF_ADVANCED